

Ratio and Proportion | Lesson-1 (Introduction) | Quantitative Aptitude

Channel: *Feel Free to Learn* [1]

1. Introduction & Foundational Importance of Quantitative Aptitude

1.1 The Four Pillars of Quantitative Aptitude

Mastering basic concepts is essential for cracking quantitative aptitude questions easily [1, 2]. The instructor highlights four core foundational topics that form the base for the entire Quantitative Aptitude section [1-3]:

1. **Percentage:** Covered across 5 to 7 dedicated lessons on the channel [1].
2. **Average:** Covered across 7 to 8 dedicated lessons on the channel [1].
3. **Number System:** A comprehensive 11-lesson series on the channel [1].
4. **Ratio and Proportion:** The current topic, planned for a complete 10-lesson series [1, 2].

Exam & Career Utility: Being strong in these four basic topics guarantees a high success rate in solving quantitative aptitude problems [1-3]. This series is specifically designed for students preparing for competitive examinations, final-year college students preparing for campus placement interviews, and candidates sitting for higher-level exams such as CAT and GATE [3, 4].

2. Core Concepts: Ratio vs. Proportion

2.1 Exam Weightage Comparison

Parameter	Ratio	Proportion
Exam Weightage	~98% of quantitative aptitude questions [4].	~2% of quantitative aptitude questions [4].
Exam Context	Forms the basis of almost all general quantitative aptitude topics [4].	Primarily appears in difficult or higher-level exams like CAT and GATE [4].
Symbolic Representation	Two dots (:) [5, 6].	Four dots (::), which means equal to (=) [6, 7].
Core Definition	Comparison of two or more items [5].	Equality of two ratios [6, 7].

2.2 Concept 1: Ratio

WHAT is a Ratio?

A **ratio** is defined as the comparison of two or more items or quantities [5]. After comparing the items and canceling common factors, the final comparison value is represented using numbers separated by two dots (:) [5, 8].

HOW is a Ratio Represented?

- **Two Items:** Represented as $a : b$ [5, 6].
- **Multiple Items:** Can compare three, four, or more items simultaneously, such as $2 : 3 : 4$ or $2 : 3 : 4 : 5$ [8].
- **Fractional Form:** The expression $a : b$ can also be written in fraction form as a / b [6]. Both representations hold the exact same mathematical meaning [6].

WHY and HOW: Instructor's Practical Example

- **Scenario:**
 - Instructor's monthly salary = ₹10,000 [5, 8].
 - Your monthly salary = ₹20,000 [5, 8].
- **Step-by-Step Ratio Calculation:**
 1. Write the initial comparison: 10,000 : 20,000 [8].
 2. Cancel out four common zeros from both terms [8].
 3. Simplify to get the integer ratio: 1 : 2 [8].
- **Interpretation:** The instructor receives 1 part of the salary, while you receive double the amount (2 parts) [8].

2.3 Concept 2: Proportion

WHAT is a Proportion?

A **proportion** represents the **equality of two ratios** [6, 7]. It compares four items or values across two distinct ratios and equates them [6, 7].

HOW is a Proportion Represented?

- Expressed symbolically using four dots (::), which directly signifies an equal sign (=) [6, 7].
- **Standard Notation:** a : b :: c : d [6, 7].
- **Equivalent Algebraic Form:** a : b = c : d or a / b = c / d [6, 7].

Practical Solved Example (Finding an Unknown Variable)

- **Given Problem:** Find the unknown value *d* if *a : b = 2 : 3* and *c : d = 4 : d* (written in proportion as 2 : 3 :: 4 : *d*) [7].
- **Step-by-Step Solution:**
 1. **Convert to fraction equality:** $\frac{2}{3} = \frac{4}{d}$ [7]
 2. **Cross-multiply to solve for *d*:** $2 \times d = 4 \times 3$ [7]
 3. **Isolate *d*:** $d = \frac{4 \times 3}{2} = \frac{12}{2} = 6$ [7, 9]
 4. **Verification / Proof:** Substitute *d* = 6 back into the right-hand side ratio: $\frac{4}{6} = \frac{2}{3}$ Since LHS = $\frac{2}{3}$ and RHS = $\frac{2}{3}$, the proportion equation is verified and balanced [7, 9].

3. Types of Ratios

The speaker details 5 specific types of ratios plus a method for calculating compound ratios [9]. These concepts frequently appear in college campus placement tests and aptitude exams [9, 10].

Ratio Type	Definition / Mathematical Rule	Instructor's Example Problem	Step-by-Step Calculation	Final Answer
1. Duplicate Ratio	Square the given ratio terms ($a^2 : b^2$) [9-11].	Find duplicate ratio of 3 : 7 [9, 10].	$3^2 : 7^2 = 9 : 49$ [10]	9 : 49 [10]
2. Triplicate Ratio	Cube the given ratio terms ($a^3 : b^3$) [9-11].	Find triplicate ratio of 3 : 7 [10].	$3^3 : 7^3 = 27 : 343$ [10]	27 : 343 [10]

Ratio Type	Definition / Mathematical Rule	Instructor's Example Problem	Step-by-Step Calculation	Final Answer
3. Sub-duplicate Ratio	Take the square root of the given terms ($\sqrt{a} : \sqrt{b}$) [9, 11, 12].	Find sub-duplicate ratio of 9 : 25 [11].	$\sqrt{9} : \sqrt{25} = 3 : 5$ [11]	3 : 5 [11]
4. Sub-triplicate Ratio	Take the cube root of the given terms ($\sqrt[3]{a} : \sqrt[3]{b}$) [9, 11, 12].	Find sub-triplicate ratio of 125 : 343 [11].	$\sqrt[3]{125} : \sqrt[3]{343} = 5 : 7$ [11]	5 : 7 [11]
5. Inverse Ratio	Reverse / invert the position of the terms ($b : a$) [9, 11, 12].	Find inverse ratio of 5 : 7 [11].	Swap terms: 5 : 7 becomes 7 : 5 [11, 12].	7 : 5 [12]

3.1 Compound Ratio Calculation

WHAT is a Compound Ratio?

A **compound ratio** is obtained by multiplying the first terms (antecedents / numerators) of all given ratios together to form the new first term, and multiplying the second terms (consequents / denominators) of all given ratios together to form the new second term [9, 12, 13].

Formula

Compound Ratio of $(a : b, c : d, e : f) = \frac{a \times c \times e}{b \times d \times f}$ or $(a \times c \times e) : (b \times d \times f)$ [12, 13]

Practical Solved Example

- Given Ratios:
 - $a : b = 1 : 3$ [12]
 - $c : d = 2 : 9$ [12]
 - $e : f = 4 : 7$ [12]
- Step-by-Step Calculation:
 - Multiply Numerators (First Terms):** Numerator = $1 \times 2 \times 4 = 8$ [12]
 - Multiply Denominators (Second Terms):** Denominator = $3 \times 9 \times 7 = 189$ [12]
 - Combine Results:** Compound Ratio = $\frac{8}{189}$ or $8 : 189$ [12]

4. The Golden Rule of Ratios: Converting Fractional Ratios

4.1 The Core Rule & Warning


IMPORTANT GOLDEN RULE: A ratio should **NEVER** be left in fraction form at any point [3, 13-16]. It must **ALWAYS** be converted into integer form [3, 14-16].

- When to apply:
 - If a question presents ratio terms as fractions, convert them to integers **before** beginning any further problem-solving [14, 16].
 - If your final calculated answer is a fractional ratio, simplify it into an integer ratio using LCM before stating the final answer [15, 16].

4.2 Standard Method to Convert Fractional Ratios to Integer Ratios

1. Identify the denominators of all fractional terms in the ratio [14, 15].
 2. Calculate the **LCM (Least Common Multiple)** of these denominators [14-16].
 3. Multiply each fractional term by the LCM (or convert all terms to have the common LCM denominator) [14, 15].
 4. Express the numerators as the final simplified integer ratio [14, 15].
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4.3 Instructor's Solved Examples

Example 1:

- **Given Fractional Ratio:** $a : b = \frac{2}{9} : \frac{1}{3}$ [14]
- **Step-by-Step Conversion:**
 1. Find LCM of denominators 9 and 3: $\text{LCM} = 9$ [14].
 2. Convert terms to common denominator 9:
 - First term: $\frac{2}{9} \rightarrow 2$ [14].
 - Second term: Multiply numerator and denominator by 3: $\frac{1 \times 3}{3 \times 3} = \frac{3}{9} \rightarrow 3$ [14].
 3. Simplified Integer Ratio: $a : b = 2 : 3$ [14].

Example 2:

- **Given Fractional Ratio:** $a : b = \frac{2}{7} : \frac{5}{14}$ [15]
 - **Step-by-Step Conversion:**
 1. Find LCM of denominators 7 and 14: $\text{LCM} = 14$ [15].
 2. Convert terms to common denominator 14:
 - First term: Multiply numerator and denominator by 2: $\frac{2 \times 2}{7 \times 2} = \frac{4}{14} \rightarrow 4$ [15].
 - Second term: $\frac{5}{14} \rightarrow 5$ [15].
 3. Simplified Integer Ratio: $a : b = 4 : 5$ [15].
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5. Summary & Lesson Roadmap

5.1 Recap of Lesson-1

Lesson 1 provided the essential theoretical foundation and basic mechanical rules for the Ratio and Proportion series [16]:

1. Basic definitions, symbolic representations ($:$, $::$), and real-world salary examples of Ratios and Proportions [16].
2. Solving basic proportion equations using cross-multiplication ($a/b = c/d$) [7, 16].
3. Understanding 5 core types of ratios (Duplicate, Triplicate, Sub-duplicate, Sub-triplicate, Inverse) and how to calculate a Compound Ratio [9, 16].
4. Converting fractional ratios into integer ratios using the LCM method [14-16].

5.2 Course Roadmap

- The instructor plans to release a total of **10 comprehensive lessons** covering the entire topic of Ratio and Proportion [2, 3].
 - Viewers are advised to watch all remaining lessons sequentially after mastering these basics [2, 3].
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COMPLETE TOPIC CHECKLIST

- ☒ **1. Introduction & Channel Context**
 - ☒ YouTube Channel: *Feel Free to Learn* [1]

- Importance of Quantitative Aptitude [1, 2]
 - Four Core Foundational Pillars: Percentage, Average, Number System, Ratio & Proportion [1-3]
 - Target Audience: Campus interviews, competitive exams, CAT, GATE [3, 4]
- 2. Concept of Ratio**
 - Exam Weightage: ~98% of Quant questions [4]
 - Definition: Comparison of two or more items [5]
 - Representation: Two dots (:) and fraction notation (a/b) [5, 6]
 - Salary Example: ₹10,000 vs ₹20,000 $\rightarrow$ 1 : 2 [5, 8]
- 3. Concept of Proportion**
 - Exam Weightage: ~2% of Quant questions (CAT / GATE) [4]
 - Definition: Equality of two ratios [6, 7]
 - Representation: Four dots (::) / Equal sign (=) [6, 7]
 - Solved Equation: $2 : 3 :: 4 : d \implies d = 6$ [7, 9]
- 4. Types of Ratios**
 - Duplicate Ratio ($a^2 : b^2$) with example $3 : 7 \rightarrow 9 : 49$ [9, 10]
 - Triplicate Ratio ($a^3 : b^3$) with example $3 : 7 \rightarrow 27 : 343$ [10]
 - Sub-duplicate Ratio ($\sqrt{a} : \sqrt{b}$) with example $9 : 25 \rightarrow 3 : 5$ [11]
 - Sub-triplicate Ratio ($\sqrt[4]{a} : \sqrt[4]{b}$) with example $125 : 343 \rightarrow 5 : 7$ [11]
 - Inverse Ratio ($b : a$) with example $5 : 7 \rightarrow 7 : 5$ [11, 12]
 - Compound Ratio ($\frac{a \times c \times e}{b \times d \times f}$) with example $1 : 3, 2 : 9, 4 : 7 \rightarrow 8 : 189$ [12, 13]
- 5. The Golden Rule of Ratios**
 - Ratios must NEVER be in fraction form [13-16]
 - LCM Conversion Method [14, 15]
 - Solved Example 1: $\frac{2}{9} : \frac{1}{3} \rightarrow 2 : 3$ [14]
 - Solved Example 2: $\frac{2}{7} : \frac{5}{14} \rightarrow 4 : 5$ [15]
- 6. Series Roadmap & Closing Notes**
 - Total 10 lessons planned for Ratio & Proportion [2, 3]

QUICK REVISION

- Foundational Rule of Quant:** Master Percentages, Averages, Number Systems, and Ratios & Proportions to easily solve any quantitative aptitude question [1-3].
- Ratio ($a : b$):** Comparison of items [5]. $a : b = \frac{a}{b}$ [6]. Represents ~98% of exam questions [4].
- Proportion ($a : b :: c : d$):** Equality of two ratios [6, 7]. Represents ~2% of exam questions [4]. $\frac{a}{b} = \frac{c}{d}$
- Types of Ratios Summary:**
 - Duplicate Ratio:** Square terms ($3 : 7 \rightarrow 9 : 49$) [10].
 - Triplicate Ratio:** Cube terms ($3 : 7 \rightarrow 27 : 343$) [10].
 - Sub-duplicate Ratio:** Square root ($\sqrt{9} : \sqrt{25} \rightarrow 3 : 5$) [11].
 - Sub-triplicate Ratio:** Cube root ($\sqrt[4]{125} : \sqrt[4]{343} \rightarrow 5 : 7$) [11].
 - Inverse Ratio:** Swap terms ($5 : 7 \rightarrow 7 : 5$) [11, 12].

◦ **Compound Ratio:** $\frac{\text{Product of antecedents}}{\text{Product of consequents}} = \frac{1 \times 2 \times 4}{3 \times 9 \times 7} = \frac{8}{189}$ [12].

- **Golden Non-Fraction Rule:** Never leave ratios as fractions [13, 14]. Always find the LCM of denominators and convert terms to integer form [14, 15].

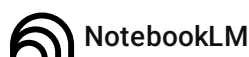
💡 Would you like me to convert these study notes into a downloadable PDF report, or generate practice quiz questions to test your understanding of these concepts?

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now lesson i have given video link do the same from now onwards no need of topic check list

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now lesson 3



Ratio and Proportion | Lesson-3(Divided Into Parts) | Quantitative Aptitude

Channel: *Feel Free to Learn*

1. Core Concept: "Divided Into Parts" Model

1.1 Overview & Competitive Exam Importance

The "**Divided Into Parts**" model is a fundamental concept in Ratio and Proportion [1].

- **Target Examinations:** Frequently appears in campus placement interviews, railway exams, SSC exams, and easy-to-moderate level competitive tests [1].
- **Higher-Level Exams:** Serves as the building block for solving complex, high-difficulty problems in CAT, GATE, and bank exams [1].
- **Core Problem Structure:** A total sum or amount (e.g., ₹10,000 or ₹8,000) is divided among two, three, or four individuals (e.g., $A : B : C$) according to a given ratio [1].
- **Common Target Questions:**
 1. Finding the specific monetary share of one person [1, 2].
 2. Finding the sum of shares received by two or more people [3].
 3. Finding the difference between the shares of two people [1, 4].
 4. Finding who gains or loses the most due to a calculation error [5].

1.2 General Formula & Method

Traditional Part-Value Formula

$$\text{Share of Person } X = \left(\frac{\text{Parts assigned to Person } X}{\text{Total Ratio Parts}} \right) \times \text{Total Amount} \quad [6]$$

Quick Unit-Value (x) Method

1. Sum all terms of the combined integer ratio to get the **Total Ratio Parts** (T) [7].
2. Set $T \times x = \text{Total Amount}$, and solve for $x = \frac{\text{Total Amount}}{T}$ (Value of 1 part) [7].
3. Multiply x by individual ratio parts to determine each person's exact share [7].

2. Fundamental Rules & Best Practices

1. **Non-Fraction Rule:** Ratio values must **never** remain in fraction form [8, 9]. Always take the LCM of denominators to convert fractional ratios into integer ratios **before** solving [9, 10].
2. **Decimal Conversion:** If a ratio contains decimals, eliminate the decimal point by dividing by powers of 10 (10, 100) and simplify to integers [11, 12].
3. **Sequential Order Verification:** Ensure ratios follow sequential connections ($A : B, B : C, C : D$) [11, 13, 14]. If given $D : C$, invert it to $C : D$ before applying the Grid Technique [11].
4. **Direct Difference Calculation:** When asked for the difference between two shares, find the difference in ratio parts first rather than calculating individual shares separately to save exam time [12, 15].

3. Detailed Solved Questions (Questions 1 to 8)

Question 1: Basic Two-Part Division

- **Problem Statement:** If ₹1,000 is divided between A and B in the ratio $3 : 2$, then A will receive how much? [2]
- **Given Data:** Total Amount = ₹1,000; Ratio $A : B = 3 : 2$ [2, 6].
- **Step-by-Step Solution:**
 1. **Calculate Total Parts:** $3 + 2 = 5$ parts [6].
 2. **Find A 's Share:** A 's Share $= \frac{3}{5} \times 1000 = 3 \times 200 = \text{₹}600$ [6]
 3. **Find B 's Share (For Verification):** B 's Share $= \frac{2}{5} \times 1000 = 2 \times 200 = \text{₹}400$ [16] Verification : $600 + 400 = \text{₹}1,000$ [16]
- **Final Answer:** ₹600 [16].

Question 2: Dividing Fractional Ratio Parts

- **Problem Statement:** If 78 is divided into three parts which are in the ratio $1 : \frac{1}{3} : \frac{1}{6}$, then find the middle part [8].
- **Given Data:** Total Value = 78; Fractional Ratio = $1 : \frac{1}{3} : \frac{1}{6}$ [8].
- **Step-by-Step Solution:**
 1. **Convert Fractional Ratio to Integer Ratio:**
 - Take the LCM of denominators 3 and 6, which is 6 [10].
 - Multiply each term by 6: $(1 \times 6) : (\frac{1}{3} \times 6) : (\frac{1}{6} \times 6) \implies 6 : 2 : 1$ [10]
 2. **Identify Middle Part & Total Ratio Parts:**
 - Middle Part = 2 ratio parts [10].
 - Total Ratio Parts = $6 + 2 + 1 = 9$ parts [10].

3. **Calculate Middle Part Share:** Middle Part = $\frac{2}{9} \times 78$ [10] Simplify by dividing numerator and denominator by 3:
 Middle Part = $\frac{2 \times 26}{3} = \frac{52}{3}$ [9, 10]

- **Final Answer:** $\frac{52}{3}$ (or 17.33) [9, 10].

Question 3: Combining Two Sub-Ratios ($A : B$ and $B : C$)

- **Problem Statement:** ₹33,630 is divided among A, B, C in such a manner that the ratio of $A : B = 3 : 7$ and $B : C = 6 : 5$. Find the amount received by person B [9, 17].
- **Given Data:** Total Amount = ₹33,630; $A : B = 3 : 7, B : C = 6 : 5$ [17].
- **Step-by-Step Solution:**
 1. **Combine Sub-Ratios to find $A : B : C$ using the Grid Method:** $\text{\texttt{\text{Row 1 } (A:B): & 3 & 7 & \text{(Fill empty) C}}}$
 $\text{\texttt{\text{ with } 7\text{(text{)}} \text{ \text{Row 2 } (B:C): & 6 & 5 & \text{(Fill empty) A \text{ with } 6\text{(text{)}} \end{array} \quad [18]}}$
 $\text{\texttt{\text{A} = 3 \times 6 = 18, \quad B = 7 \times 6 = 42, \quad C = 7 \times 5 = 35 \quad [18]}}$
 $\text{\texttt{\text{A} : B : C = 18 : 42 : 35 \quad [18]}}$
 2. **Calculate Total Parts:** Total Parts = $18 + 42 + 35 = 95$ [18]
 3. **Calculate Person B 's Share:** B 's Share = $\left(\frac{42}{95}\right) \times 33630$ [18] $\frac{33630}{95} = 354$ [19] B 's Share = $42 \times 354 = ₹14,868$ [19]
- **Final Answer:** ₹14,868 (Instructor note: written as 14,868 / 13,868 during board multiplication) [19].

Question 4: Four-Part Ratio Combination ($A : B : C : D$) & Sum of Shares

- **Problem Statement:** ₹3,400 is divided among four persons A, B, C, D such that $A : B = 2 : 3, B : C = 4 : 3$, and $C : D = 2 : 3$. Find the sum of the shares of B and D [3].
- **Given Data:** Total Amount = ₹3,400; $A : B = 2 : 3, B : C = 4 : 3, C : D = 2 : 3$ [3, 13].
- **Step-by-Step Solution:**
 1. **Combine to find $A : B : C : D$ via Grid Method:** $\text{\texttt{\text{Row 1): & 2 & 3 & 3 & 3 \text{ \text{Row 2): & 4 & 4 & 3 & 3 \text{ \text{Row 3): & 2 & 2 & 2 & 3 \end{array} \quad [13]}}$
 $\text{\texttt{\text{A} = 2 \times 4 \times 2 = 16 \quad [13]}}$
 $\text{\texttt{\text{B} = 3 \times 4 \times 2 = 24 \quad [13]}}$
 $\text{\texttt{\text{C} = 3 \times 3 \times 2 = 18 \quad [13]}}$
 $\text{\texttt{\text{D} = 3 \times 3 \times 3 = 27 \quad [13]}}$
 $\text{\texttt{\text{A} : B : C : D = 16 : 24 : 18 : 27 \quad [13, 20]}}$
 2. **Find Combined Target Parts for $B + D$:** Parts for $(B + D) = 24 + 27 = 51$ parts [20] Total Parts = $16 + 24 + 18 + 27 = 85$ parts [20]
 3. **Calculate Total Amount Received by B and D :** Sum of $(B + D) = \left(\frac{51}{85}\right) \times 3400$ [20] $\frac{3400}{85} = 40$ [20]
 Sum of $(B + D) = 51 \times 40 = ₹2,040$ [20]
- **Final Answer:** ₹2,040 [20].

Question 5: Erroneous Division Analysis (Correct vs. Wrong Ratio)

- **Problem Statement:** By mistake, instead of dividing ₹117 among A, B, C in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$, it was divided in the ratio $2 : 3 : 4$. Who gains the most, and by how much? [5, 20]
- **Given Data:** Total Amount = ₹117 [5].
 - Intended Ratio = $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ [5].
 - Wrongly Applied Ratio = $2 : 3 : 4$ [5].
- **Step-by-Step Solution:**
 1. **Correct Division Calculation:**

- Convert fractional ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ using LCM (12): $(\frac{1}{2} \times 12) : (\frac{1}{3} \times 12) : (\frac{1}{4} \times 12) \implies 6 : 4 : 3$ [21]
- Sum of parts = $6 + 4 + 3 = 13$ parts [7].
- Value per part (x) = $\frac{117}{13} = ₹9$ [7].
- Actual intended shares:
 - $A = 6 \times 9 = ₹54$ [7]
 - $B = 4 \times 9 = ₹36$ [7]
 - $C = 3 \times 9 = ₹27$ [7]

2. Wrong Division Calculation:

- Wrong Ratio = $2 : 3 : 4$ [22].
- Sum of parts = $2 + 3 + 4 = 9$ parts [22].
- Value per part (x) = $\frac{117}{9} = ₹13$ [22].
- Shares received in wrong division:
 - $A = 2 \times 13 = ₹26$ [22]
 - $B = 3 \times 13 = ₹39$ [22]
 - $C = 4 \times 13 = ₹52$ [22]

3. Gain/Loss Comparison Table:

Person	Intended Share (₹)	Received Share (₹)	Difference (₹)	Result
A	₹54	₹26	−28	Loss of ₹28 [22]
B	₹36	₹39	+3	Gain of ₹3 [22]
C	₹27	₹52	+25	Gain of ₹25 [22]

- **Final Answer:** Person C gains the most, gaining ₹25 [23].

Question 6: Word Statements to Ratio Conversion

- **Problem Statement:** If a sum of money is divided among A, B, C such that A 's share is equal to twice B 's share, and B 's share is 4 times C 's share, find the ratio of their shares ($A : B : C$) [24].
- **Step-by-Step Solution:**
 - 1. Translate Word Statements into Ratios:**
 - A gets twice B 's share $\implies A = 2B \implies A : B = 2 : 1$ [24, 25].
 - B gets 4 times C 's share $\implies B = 4C \implies B : C = 4 : 1$ [24, 25].
 - 2. Combine via Grid Method:** $\text{Row 1 } (A:B): 2 \& 1 \& 1 \quad \text{Row 2 } (B:C): 4 \& 4 \& 1 \quad \text{[25]}$
 $\$A = 2 \times 4 = 8, \quad B = 1 \times 4 = 4, \quad C = 1 \times 1 = 1 \quad \text{[25]}$
- **Final Answer:** $8 : 4 : 1$ [25].

Question 7: Fraction Equations and Multi-Variable Distribution

- **Problem Statement:** Divide ₹1,250 among A, B, C so that A gets $\frac{2}{9}$ of B 's share and C gets $\frac{3}{4}$ of A 's share. Find the individual shares of A, B , and C [26].
- **Given Data:** Total Amount = ₹1,250 [26]; $A = \frac{2}{9}B$ [26], $C = \frac{3}{4}A$ [26].

• **Step-by-Step Solution:**

1. **Express Sub-Ratios:**

- $A = \frac{2}{9}B \implies \frac{A}{B} = \frac{2}{9} \implies A : B = 2 : 9$ [14].
- Substitute $A = \frac{2}{9}B$ into $C = \frac{3}{4}A$: $C = \frac{3}{4} \times (\frac{2}{9}B) = \frac{1}{6}B \implies \frac{C}{B} = \frac{1}{6} \implies B : C = 6 : 1$ [14]

2. **Combine to find $A : B : C$:** $\text{Row 1: } 2 \text{ \& } 9 \text{ \& } 9 \text{ \& } \text{Row 2: } 6 \text{ \& } 6 \text{ \& } 1 \text{ \& } \quad [27] \$\$ \$\$ A = 2 \times 6 = 12, \quad B = 9 \times 6 = 54, \quad C = 9 \times 1 = 9 \quad [27] \$\$ Simplify by dividing all terms by \3: \$\$ A : B : C = 4 : 18 : 3 \quad [27] \$\$$

3. **Calculate Individual Shares:** Total Parts = $4 + 18 + 3 = 25$ parts [27] Value of 1 Part (x) = $\frac{1250}{25} = ₹50$ [27]

- A 's Share = $4 \times 50 = ₹200$ [27, 28]
- B 's Share = $18 \times 50 = ₹900$ [28]
- C 's Share = $3 \times 50 = ₹150$ [27, 28]

• **Final Answer:** $A = ₹200, B = ₹900, C = ₹150$ [28].

Question 8: Decimal Ratio Conversion & Exceeding Share Calculation

- **Problem Statement:** ₹2,010 is divided among A, B, C such that if A gets ₹5, B must get ₹12; if B gets ₹4, C gets ₹5.50. By how much does C 's share exceed B 's share? [29]

- **Given Data:** Total Amount = ₹2,010 [29].

- $A : B = 5 : 12$ [11, 29].
- $B : C = 4 : 5.50$ [11].

• **Step-by-Step Solution:**

1. **Convert Decimal Ratio $B : C$ to Integer:** $B : C = 4 : \frac{550}{100} = 4 : \frac{11}{2} \implies (4 \times 2) : 11 = 8 : 11$ [11]

2. **Combine to find $A : B : C$:** $\text{Row 1: } 5 \text{ \& } 12 \text{ \& } 12 \text{ \& } \text{Row 2: } 8 \text{ \& } 8 \text{ \& } 11 \text{ \& } \quad [4] \$\$ \$\$ A = 5 \times 8 = 40, \quad B = 12 \times 8 = 96, \quad C = 12 \times 11 = 132 \quad [4] \$\$ Simplify by dividing terms by \4: \$\$ A : B : C = 10 : 24 : 33 \quad [4] \$\$$

3. **Calculate Direct Difference (Exceeding Amount):**

- Difference in parts between C and $B = 33 - 24 = 9$ parts [4, 15].
- Total Ratio Parts = $10 + 24 + 33 = 67$ parts [4]. Exceeding Amount = $(\frac{9}{67}) \times 2010$ [4, 12] $\frac{2010}{67} = 30$ [12] Exceeding Amount = $9 \times 30 = ₹270$ [12]

- **Final Answer:** C 's share exceeds B 's share by **₹270** [12, 15].

4. Practice Homework Assignments

The instructor assigned two homework practice questions at the conclusion of Lesson 3 [15]:

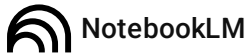
1. **Homework Task 1:** Practice problems on dividing amounts with fractional ratios after converting to integer ratios via LCM [2, 15].
2. **Homework Task 2:** Practice problems on calculating exceeded shares and differences directly using ratio part subtraction [12, 15].

QUICK REVISION

- **Primary Part Formula:** Share = $\left(\frac{\text{Target Ratio Part}}{\text{Sum of All Ratio Parts}} \right) \times \text{Total Amount}$ [6].

- **Unit Part Method (x):** Find $x = \frac{\text{Total Amount}}{\text{Total Ratio Parts}}$, then multiply by individual terms [7, 27].
- **Fraction Rule:** Convert fractional ratios using LCM of denominators before calculating [10, 21].
- **Decimal Rule:** Express decimals as fractions ($\frac{\text{Value}}{100}$), cancel common factors, and cross-multiply to get integer ratios [11].
- **Direct Difference Shortcut:** Difference Amount = $\left(\frac{\text{Difference in Ratio Parts}}{\text{Total Ratio Parts}} \right) \times \text{Total Amount}$ [4, 12].

💡 Would you like me to convert these Lesson 3 study notes into an editable file, create a study guide PDF report, or generate practice quiz questions to test your speed and accuracy?



Ratio and Proportion | Lesson-4(Based On Numbers) | Quantitative Aptitude

Channel: *Feel Free to Learn* [1]

1. Overview, Exam Scope, and Core Foundational Rules

1.1 Importance & Exam Utility

- **Target Audience & Exams:** This lesson focuses on ratio problems specifically based on numbers [1]. Questions from this concept appear primarily in **easy-level competitive examinations**, campus placement interviews, and IT sector quantitative aptitude rounds [2, 3].
- **Higher-Level Exam Relevance:** While candidates appearing for bank exams, CAT, GATE, or other tough/medium-level competitive tests may not get direct, basic number questions [2, 3], mastering these foundational concepts is mandatory [2]. They serve as essential building blocks for upcoming complex topics such as **ages**, **coins**, and **partnership** [4].

1.2 Core Rule: Variable (x) Assignment

⚠️ **THE GOLDEN VARIABLE RULE:** Whenever a question provides values in terms of a ratio and asks for the final answer in terms of **actual numbers**, you **must** assign a variable (e.g., $3x$ and $8x$) to the terms [3, 5, 6]. Never use raw ratio values directly in algebraic equations [3, 6, 7].

- **When Ratio is Given:** Assume terms as ax, bx, cx [3, 6, 8].
- **When Ratio is NOT Given:** Assume the unknown numbers as separate variables x and y , set up simultaneous equations, and solve for x and y before establishing their ratio [6, 9].

2. Detailed Solved Questions (Questions 1 to 7)

Question 1: Finding the Smallest Number from a Ratio & Difference

- **Problem Statement:** The ratio of two numbers is $3 : 8$ and their difference is 115. Find the smallest of the two numbers [3].
- **Given Data:** Ratio = $3 : 8$; Difference = 115 [3].
- **Step-by-Step Solution:**
 1. **Assign Variables:** Let the first number be $3x$ and the second number be $8x$ [3, 10].
 2. **Identify Target:** The smallest term between $3x$ and $8x$ is $3x$ [10].
 3. **Set up Equation using Difference:** $8x - 3x = 115$ [10] $5x = 115 \implies x = \frac{115}{5} = 23$ [10]
 4. **Calculate Smallest Number:** Smallest Number = $3x = 3 \times 23 = 69$ [5, 10]

- **Instructor's Extra Tips:**
 - **Largest Number:** Substitute $x = 23$ into $8x \implies 8 \times 23 = 184$ [5].
 - **Sum of Numbers:** Either add $69 + 184 = 253$ or evaluate $3x + 8x = 11x \implies 11 \times 23 = 253$ [5].
 - **Final Answer:** 69 [5].
-

Question 2: Finding the Sum of Two Numbers

- **Problem Statement:** The ratio of two numbers is $10 : 7$ and their difference is 105. Find the sum of the two numbers [7].
 - **Given Data:** Ratio = $10 : 7$; Difference = 105 [7].
 - **Step-by-Step Solution:**
 1. **Assign Variables:** Let the numbers be $10x$ and $7x$ [7].
 2. **Set up Difference Equation:** $10x - 7x = 105$ [7] $3x = 105 \implies x = 35$ [7]
 3. **Calculate Sum (Direct Method):** Sum = $10x + 7x = 17x$ [7, 11] Sum = $17 \times 35 = 595$ [11]
 4. **Alternative Method (Individual Values):** First Number = $10 \times 35 = 350$ [7, 11] Second Number = $7 \times 35 = 245$ [7, 11] Sum = $350 + 245 = 595$ [11]
 - **Final Answer:** 595 [11].
-

Question 3: Determining Ratio from Sum and Difference

- **Problem Statement:** The sum of two numbers is 40 and the difference of the two numbers is 4. Find the ratio of the two numbers [9, 11].
 - **Given Data:** Sum = 40; Difference = 4 [9, 11].
 - **Concept:** Since no initial ratio is provided, assign independent variables x and y [6, 9].
 - **Step-by-Step Solution:**
 1. **Form Simultaneous Equations:** Equation 1: $x + y = 40$ [9] Equation 2: $x - y = 4$ [9]
 2. **Solve for x by adding Equation 1 and Equation 2:** $(x + y) + (x - y) = 40 + 4$ [9] $2x = 44 \implies x = 22$ [9]
 3. **Solve for y :** $22 + y = 40 \implies y = 40 - 22 = 18$ [6]
 4. **Form & Simplify Ratio:** Ratio = $x : y = 22 : 18$ [6] Divide both terms by 2: Simplified Ratio = $11 : 9$ [6]
 - **Final Answer:** $11 : 9$ [6].
-

Question 4: Sum of Squares of Three Numbers

- **Problem Statement:** Three numbers are in the ratio $3 : 2 : 5$ and the sum of their squares is 1862. Find the smallest number [8].
 - **Given Data:** Ratio = $3 : 2 : 5$; Sum of Squares = 1862 [8].
 - **Step-by-Step Solution:**
 1. **Assign Variables:** Numbers are $3x$, $2x$, and $5x$ [8]. Smallest term is $2x$ [8, 12].
 2. **Set up Square Sum Equation:** $(3x)^2 + (2x)^2 + (5x)^2 = 1862$ [8, 12] $9x^2 + 4x^2 + 25x^2 = 1862$ [12] $38x^2 = 1862$ [12]
 3. **Solve for x :** $x^2 = \frac{1862}{38} = 49$ [12] $x = \sqrt{49} = 7$ [12, 13]
 4. **Calculate Smallest Number:** Smallest Number = $2x = 2 \times 7 = 14$ [12, 13]
 - **Final Answer:** 14 [13].
-

Question 5: Product of First and Third Terms (Combining Ratios)

- **Problem Statement:** Of three numbers, the ratio of the first and second number is $8 : 9$ and that of the second and third number is $3 : 4$. If the product of the first and third numbers is 2400, find the second number [13, 14].
- **Given Data:** $1^{\text{st}} : 2^{\text{nd}} = 8 : 9$; $2^{\text{nd}} : 3^{\text{rd}} = 3 : 4$; Product of $1^{\text{st}} \times 3^{\text{rd}} = 2400$ [14, 15].
- **Step-by-Step Solution:**
 1. **Combine Sub-ratios using Grid Method to find $1^{\text{st}} : 2^{\text{nd}} : 3^{\text{rd}}$:** $\begin{array}{l} \text{Row 1 } (1^{\text{st}}:2^{\text{nd}}): 8 \text{ \& } 9 \\ \text{Row 2 } (2^{\text{nd}}:3^{\text{rd}}): 3 \text{ \& } 4 \end{array}$ [14, 15] $\text{Unimplified Products: } (8 \times 3) : (9 \times 3) : (9 \times 4) = 24 : 27 : 36$ [15] $\text{Divide across all terms by 3: } 8 : 9 : 12$ [15]
 2. **Assign Variables:** First = $8x$, Second = $9x$, Third = $12x$ [15, 16].
 3. **Set up Product Equation:** $8x \times 12x = 2400$ [15] $96x^2 = 2400$ [15] $x^2 = \frac{2400}{96} = 25 \implies x = \sqrt{25} = 5$ [15, 16]
 4. **Calculate Second Number:** Second Number = $9x = 9 \times 5 = 45$ [15, 16]
- **Final Answer:** 45 [16].

Question 6: Fractional Ratios & Difference Between Extremes

- **Problem Statement:** Three numbers are in the ratio $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$. The difference between the greatest and the smallest number is 36. Find the three numbers [17, 18].
- **Given Data:** Fractional Ratio = $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$; Difference (Greatest – Smallest) = 36 [17, 18].
- **Step-by-Step Solution:**
 1. **Convert Fractional Ratio to Integers:**
 - Take LCM of denominators $(2, 3, 4) = 12$ [17].
 - Multiply each fraction term by 12: $(\frac{1}{2} \times 12) : (\frac{2}{3} \times 12) : (\frac{3}{4} \times 12) \implies 6 : 8 : 9$ [17, 18]
 2. **Identify Extremes:** Smallest term = $6x$, Greatest term = $9x$ [18].
 3. **Set up Difference Equation:** $9x - 6x = 36$ [18] $3x = 36 \implies x = 12$ [18]
 4. **Calculate All Three Numbers:**
 - First Number = $6x = 6 \times 12 = 72$ [18]
 - Second Number = $8x = 8 \times 12 = 96$ [18]
 - Third Number = $9x = 9 \times 12 = 108$ [18]
- **Instructor Warning:** Avoid working directly with fractional ratio terms [19]. Always convert fractions to integer ratios first using LCM to prevent common numerator/denominator calculation mistakes [19].
- **Final Answer:** 72, 96, and 108 [18, 19].

Question 7: Percentage Relations on Ratio Terms

- **Problem Statement:** Two numbers are in the ratio $5 : 4$. If 40% of the first number is 12, find 50% of the second number [19, 20].
- **Given Data:** Ratio = $5 : 4$; 40% of 1st Number = 12 [19, 20].
- **Step-by-Step Solution:**
 1. **Assign Variables:** First number = $5x$, Second number = $4x$ [20].
 2. **Set up Percentage Equation for First Number:** 40% of $5x = 12$ [20] $\frac{40}{100} \times 5x = 12$ [20] $\frac{2}{5} \times 5x = 12 \implies 2x = 12 \implies x = 6$ [20]

3. **Calculate Second Number:** $\text{Second Number} = 4x = 4 \times 6 = 24$ [20, 21]

4. **Find 50% of Second Number:** $50\% \text{ of } 24 = \frac{1}{2} \times 24 = 12$ [21]

- **Final Answer:** 12 [21].

3. Practice Homework Assignments

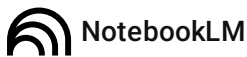
The instructor assigned self-practice problems covering:

1. Solving problems involving differences and sums of terms represented as quadratic variables (x^2) [12, 15].
2. Practice converting fractional ratios to integer ratios before evaluating percentage constraints [17, 19, 20].

QUICK REVISION

- **Variable Assignment Rule:** When ratio is given and exact numbers are asked, use ax, bx, cx [3, 6, 7]. When ratio is NOT given, use x and y equations [6, 9].
- **Sum of Squares Pattern:** $(ax)^2 + (bx)^2 + (cx)^2 = k \implies (a^2 + b^2 + c^2)x^2 = k$ [8, 12] Solve for $x = \sqrt{\frac{k}{a^2 + b^2 + c^2}}$ [12].
- **Product Pattern:** $(ax) \times (cx) = P \implies (a \cdot c)x^2 = P$ [15] Solve for $x = \sqrt{\frac{P}{a \cdot c}}$ [15].
- **Fractional Ratio Rule:** Never keep fractions; multiply by LCM of denominators first [17, 19].
- **Percentage Shortcut:** 50% of a number means taking half ($\frac{1}{2}$) of that value [21].

💡 Would you like me to generate flashcards, practice quiz questions, or a consolidated study guide combining all 4 lessons covered so far?



Ratio and Proportion | Lesson-5(Boys and Girls) | Quantitative Aptitude

Channel: *Feel Free to Learn*

1. Introduction & Competitive Exam Relevance

1.1 Overview of the "Boys and Girls in a Classroom" Concept

Lesson 5 focuses specifically on quantitative aptitude problems based on the "**Boys and Girls**" classroom/school/college model.

- **Cross-Topic Application:** This model is not restricted to Ratio and Proportion alone. The exact same concepts and techniques appear across multiple Quantitative Aptitude topics, including:
 1. **Time and Work**
 2. **Time and Distance**
 3. **Data Interpretation (DI)** (e.g., Pie charts, bar graphs, or tables where 5 to 6 questions frequently center around male/female or boy/girl ratios).
- **Target Examinations:** Crucial for all competitive examinations, campus placement aptitude tests, bank exams, SSC, Railway exams, CAT, and GATE.

2. Fundamental Rules & Solving Framework

2.1 The Two Primary Case Scenarios

When solving classroom ratio problems, determine which of the following two scenarios applies:

Parameter	Scenario A: Total Strength IS Given	Scenario B: Total Strength IS NOT Given
Initial Data	Overall number of students (e.g., 504 or 660) + Initial Ratio ($a : b$).	Only ratios (e.g., 5 : 7) + numerical additions/subtractions.
First Step	Find actual numerical count of boys and girls using parts formula.	Assign algebraic variables: Boys = ax , Girls = bx .
Equation Setup	Add/subtract numbers directly to the actual counts: $\frac{\text{Boys} \pm \text{Change}}{\text{Girls} \pm \text{Change}} = \text{New Ratio}$.	Set up variable fraction equation: $\frac{ax \pm \text{Change}}{bx \pm \text{Change}} = \text{New Ratio}$.
Goal	Simplify the resulting numerical ratio.	Cross-multiply to solve for x , then find target values.

2.2 Critical Warning & Common Student Mistake

 **CRITICAL WARNING:** Never add or subtract actual student numbers directly to or from ratio terms!

- Incorrect:** If the ratio of boys to girls is 13 : 11 and 12 girls are added, doing $11 + 12 = 23$ to claim a new ratio of 13 : 23 is **completely wrong**. Ratios represent relative parts, not absolute counts.
- Correct:** Always convert ratio terms to actual student counts (or assign x -variables) **before** adding or subtracting numerical student quantities.

2.3 Solving Multi-Ratio Problems (3 or 4 Ratio Terms)

When a problem involves 3 or 4 ratio terms (e.g., Class 1, Class 2, Class 3 with ratio 2 : 3 : 5):

- Traditional Method:** Select **any two** ratio terms to set up a standard two-variable fraction equation (e.g., $\frac{\text{Class 1}}{\text{Class 2}} = \frac{\text{New Ratio 1}}{\text{New Ratio 2}}$) and solve for x .
- Shortcut Method (Equal Increase Rule):** If an **equal number** of students is added to or subtracted from every class, check the increase in ratio parts across all terms. If the ratio part increase is uniform, equate the part difference directly to the added number of students.
 -  **Shortcut Limitation:** The shortcut works **only** when the exact same number of students is added/subtracted across all classes. If different numbers of students are added (e.g., +40, +30, +90), you **must** use the Traditional Method.

3. Detailed Solved Questions (Questions 1 to 7)

Question 1: Total Strength Given & Finding New Ratio

- Problem Statement:** The ratio of the number of boys to girls in a school with 504 students is 13 : 11. What will be the new ratio if 12 more girls are admitted?
- Given Data:** Total Students = 504; Initial Ratio (Boys : Girls) = 13 : 11; Added Girls = 12.
- Step-by-Step Solution:**
 - Calculate Total Ratio Parts:** Total Parts = $13 + 11 = 24$ parts

2. **Calculate Initial Number of Boys:** Boys = $\left(\frac{13}{24}\right) \times 504 = 13 \times 21 = 273$

3. **Calculate Initial Number of Girls:** Girls = $504 - 273 = 231$ or $\left(\frac{11}{24}\right) \times 504 = 231$

4. **Add 12 Girls to Initial Girls:** New Girls Count = $231 + 12 = 243$

5. **Form & Simplify New Ratio:** New Ratio (Boys : Girls) = $273 : 243$ Divide both terms by 3: $\frac{273}{3} : \frac{243}{3} = 91 : 81$

- **Final Answer:** 91 : 81.
-

Question 2: Total Strength Not Given & Difference Calculation

- **Problem Statement:** In a classroom, the boys and girls are in the ratio 5 : 7. If 9 students are subtracted from each of them, the ratio becomes 7 : 11. Find the difference between the number of boys and girls.
 - **Given Data:** Initial Ratio = 5 : 7; 9 subtracted from boys and 9 subtracted from girls; New Ratio = 7 : 11.
 - **Step-by-Step Solution:**
 1. **Assign Variables:** Let Boys = $5x$, Girls = $7x$.
 2. **Set up Fraction Equation:** $\frac{5x-9}{7x-9} = \frac{7}{11}$
 3. **Cross-Multiply & Solve for x :** $11(5x - 9) = 7(7x - 9)$ $55x - 99 = 49x - 63$ $55x - 49x = 99 - 63$ $6x = 36 \implies x = 6$
 4. **Calculate Difference:**
 - **Method A (Individual Values):** Boys = $5 \times 6 = 30$, Girls = $7 \times 6 = 42$ Difference = $42 - 30 = 12$
 - **Method B (Shortcut on Parts):** Difference in parts = $7x - 5x = 2x$ Difference = $2 \times 6 = 12$
 - **Final Answer:** 12.
-

Question 3: Three Classes & Traditional vs. Shortcut Comparison

- **Problem Statement:** The students in three classes are in the ratio 2 : 3 : 5. If 40 students are increased in each class, the ratio becomes 4 : 5 : 7. Find the original total number of students.
- **Given Data:** Initial Ratio = 2 : 3 : 5; Added Students per Class = 40; New Ratio = 4 : 5 : 7.

Method 1: Traditional Method

1. Assign variables: Class 1 = $2x$, Class 2 = $3x$, Class 3 = $5x$. Total Original Strength = $2x + 3x + 5x = 10x$
2. Select any two classes (e.g., Class 1 and Class 2) to set up the equation: $\frac{2x+40}{3x+40} = \frac{4}{5}$
3. Cross-multiply to solve for x : $5(2x + 40) = 4(3x + 40)$ $10x + 200 = 12x + 160 \implies 2x = 40 \implies x = 20$
4. Calculate Total Original Strength: Total Strength = $10x = 10 \times 20 = 200$

Method 2: Shortcut Method (Equal Increase Check)

1. Compare ratio parts across all three classes:
 - Class 1: From 2 parts to 4 parts $\rightarrow$ Increased by 2 parts.
 - Class 2: From 3 parts to 5 parts $\rightarrow$ Increased by 2 parts.
 - Class 3: From 5 parts to 7 parts $\rightarrow$ Increased by 2 parts.
 2. Since all classes show a uniform increase of 2 parts due to adding 40 students: $2 \text{ parts} = 40 \text{ students} \implies 1 \text{ part} = 20 \text{ students}$
 3. Total original ratio parts = $2 + 3 + 5 = 10$ parts. Total Original Strength = $10 \times 20 = 200$
- **Final Answer:** 200.
-

Question 4: Opposite Actions (Boys Leaving & Girls Joining)

- **Problem Statement:** The ratio of boys and girls in a college is 5 : 3. If 50 boys leave the college and 50 girls join the college, the ratio becomes 9 : 7. Find the original number of boys in the college.
 - **Given Data:** Initial Ratio = 5 : 3; Boys leaving = 50; Girls joining = 50; New Ratio = 9 : 7.
 - **Step-by-Step Solution:**
 1. **Assign Variables:** Let Boys = $5x$, Girls = $3x$.
 2. **Set up Equation:** $\frac{5x-50}{3x+50} = \frac{9}{7}$
 3. **Cross-Multiply & Solve for x :** $7(5x - 50) = 9(3x + 50)$ $35x - 350 = 27x + 450$ $35x - 27x = 450 + 350$ $8x = 800 \implies x = 100$
 4. **Calculate Original Number of Boys:** Boys = $5x = 5 \times 100 = 500$ (Note: Original Girls = $3x = 3 \times 100 = 300$)
 - **Final Answer:** 500.
-

Question 5: Two Sequential Stages of Students Leaving

- **Problem Statement:** The ratio of the number of boys to girls in a group becomes 2 : 1 when 15 girls leave. Afterwards, when 45 boys also leave, the ratio becomes 1 : 5. Find the original number of girls in the group.
 - **Given Data:**
 - Stage 1: 15 girls leave $\implies$ Ratio (Boys : Girls) = 2 : 1.
 - Stage 2: 45 boys also leave $\implies$ Ratio (Boys : Girls) = 1 : 5.
 - **Step-by-Step Solution:**
 1. **Define Initial Unknowns:** Let original Boys = x , original Girls = y .
 2. **Process Stage 1 Data:** $\frac{x}{y-15} = \frac{2}{1} \implies x = 2(y - 15) \implies x = 2y - 30$ — (Equation 1)
 3. **Process Stage 2 Data:** $\frac{x-45}{y-15} = \frac{1}{5}$ $5(x - 45) = 1(y - 15)$ $5x - 225 = y - 15 \implies 5x = y + 210$ — (Equation 2)
 4. **Substitute Equation 1 into Equation 2:** $5(2y - 30) = y + 210$ $10y - 150 = y + 210$ $10y - y = 210 + 150$ $9y = 360 \implies y = 40$
 - **Final Answer:** 40.
-

Question 6: Total Strength Given + Mixed Addition & Subtraction

- **Problem Statement:** The total number of students in a school was 660. The ratio between boys and girls was 13 : 9. After some days, 30 girls joined the school and some boys left the school. The new ratio of boys and girls becomes 6 : 5. Find the number of boys who left the school.
 - **Given Data:** Total Strength = 660; Initial Ratio = 13 : 9; Girls added = 30; Boys left = x ; New Ratio = 6 : 5.
 - **Step-by-Step Solution:**
 1. **Calculate Initial Boys and Girls Counts:** Total Parts = $13 + 9 = 22$ parts Initial Boys = $\left(\frac{13}{22}\right) \times 660 = 13 \times 30 = 390$ Initial Girls = $660 - 390 = 270$
 2. **Apply Changes:** New Girls Count = $270 + 30 = 300$ New Boys Count = $390 - x$
 3. **Set up & Solve Ratio Equation:** $\frac{390-x}{300} = \frac{6}{5}$ Cancel denominator 300 by 5: $\frac{390-x}{60} = \frac{6}{5}$ $390 - x = 60 \times 6 = 360$ $x = 390 - 360 = 30$
 - **Final Answer:** 30.
-

Question 7: Total Strength Given + Finding Required Girls to Add

- **Problem Statement:** In a college union, there are 48 students. The ratio of the number of boys to girls is 5 : 3. How many girls must be added to the union so that the ratio of boys to girls becomes 6 : 5?
 - **Given Data:** Total Union Strength = 48; Initial Ratio = 5 : 3; Target Ratio = 6 : 5.
 - **Step-by-Step Solution:**
 1. **Calculate Initial Counts:** Total Parts = 5 + 3 = 8 parts Boys = $\left(\frac{5}{8}\right) \times 48 = 30$ Girls = 48 - 30 = 18
 2. **Set up Equation with Added Girls (x):** $\frac{30}{18+x} = \frac{6}{5}$
 3. **Cross-Multiply & Solve for x:** $30 \times 5 = 6(18 + x)$ $150 = 108 + 6x$ $6x = 150 - 108 = 42 \implies x = 7$
 - **Final Answer:** 7.
-

4. Practice Homework Assignments

The instructor assigned 3 self-practice homework problems at the end of Lesson 5 covering:

1. Calculating new classroom ratios after numerical additions to a given total strength.
 2. Solving multi-class ratio increase problems using both traditional and shortcut techniques.
 3. Solving multi-stage student departure/entry equations to determine initial group sizes.
-

QUICK REVISION

- **Total Strength Given:** Calculate actual initial boys/girls counts first using $\left(\frac{\text{Part}}{\text{Total Parts}}\right) \times \text{Total}$, then apply changes and simplify.
- **Total Strength Not Given:** Assign variables (ax, bx), build fraction $\frac{ax \pm \text{change}}{bx \pm \text{change}} = \frac{\text{New 1}}{\text{New 2}}$, and solve for x .
- **Never Add Numbers to Ratios:** Do not add/subtract actual student counts directly to ratio terms.
- **Equal Increase Shortcut (3+ Classes):** If every class receives the exact same number of added students, verify if ratio parts increase by a uniform amount k . Then set k parts = Added Students to find 1 part.
- **Two-Stage Equation Strategy:** In multi-step departure problems, construct Equation 1 from Stage 1 ($x = 2y - 30$) and substitute directly into Stage 2 equation.